Control Flow

Name: Rahul Vedant Pemsing
Student ID: 36305529
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Learning Journey and Evidence

Prior Knowledge and Experience

I am very familiar with control flow statements from different programming languages namely, Visual basic(console) and python that i did in the past but was completely ignorant to some other:
The one i was unfamiliar with are as follows:

- The key presses examples mentioned in the field guide(AnyKeyPressed,KeyUp,KeyDown,KeyReleased,KeyTyped,MouseClicked,MouseDown,MouseUp),
- Loops-> The CONTINUE and GOTO statements.

Study to Acquire, Refresh, and/or Extend Capability

To study and master this topic I did the following:

01/08/2025 - Went over the syntax for control flow statement in C#
I learn what the equivalent syntax for the statement was in c# compared to visual basics.
I was able to learn the syntax and was ready to start coding.

01/08/2025 - Coded simple stats; test your knowledge

Program.cs > Program > <top-level-statements-entry-point>

```
string tempval, choice;
using static SplashKitSDK.SplashKit;
    }

    total += ConvertToDouble(tempval);
    count += 1;
    tempval = "";

    WriteLine($"Count: {count}");
    WriteLine($"Total: {total}");
    WriteLine($"Min: {min}");
    WriteLine($"Max: {max}");
    WriteLine($"Average: {total / count}");
    WriteLine();

    Write("Add another value: [y/n] ");
    choice = ReadLine();

} while (choice == "y");
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
Min: -45
Max: 30
Average: 0

Add another value: [y/n] y
Enter value: 25.5
Count: 5
Total: 25.5
Min: -45
Max: 30
Average: 5.1

Add another value: [y/n] n
I hope you got the information you are after!

Vedant@LAPTOP-Q7B93650 MINGW64 /c/Users/Vedant/Desktop/mycode/my_c#_project/simple_stats
$
```

```
string tempval, choice;
using static SplashKitSDK.SplashKit;



do
{
    Write("Enter value: ");
    tempval = ReadLine();
    val = ConvertToDouble(tempval);
    if (val < min) ;

    {
        min = val;
    }
```

Ran into a dumb mistake, due to mis-typing “;” at end of one of the if condition lines rendering it useless. I had only realised that mis-take after running the program and my value of min wasn’t changing. Thankfully it was an easy fix of just removing the “;”.
Additionally from this program, I learned the proper syntax for control flow for C#.

02/08/2025 - Coded simple music plsyer; test your knowledge

```
Program.cs
Program.cs > Program > <top-level-statements-entry-point>
string opt, song, song_path, song_to_play;
using static SplashKitSDK.SplashKit;
            Write("What is the song name: ");
            song_to_play = ReadLine();

            if (song_to_play != song)
            {
                WriteLine($"There is nothing called {song_to_play} loaded");
                WriteLine();
                break;
            }
            else
            {
                PlayMusic(song);
                flag = true;
                break;

            }
```

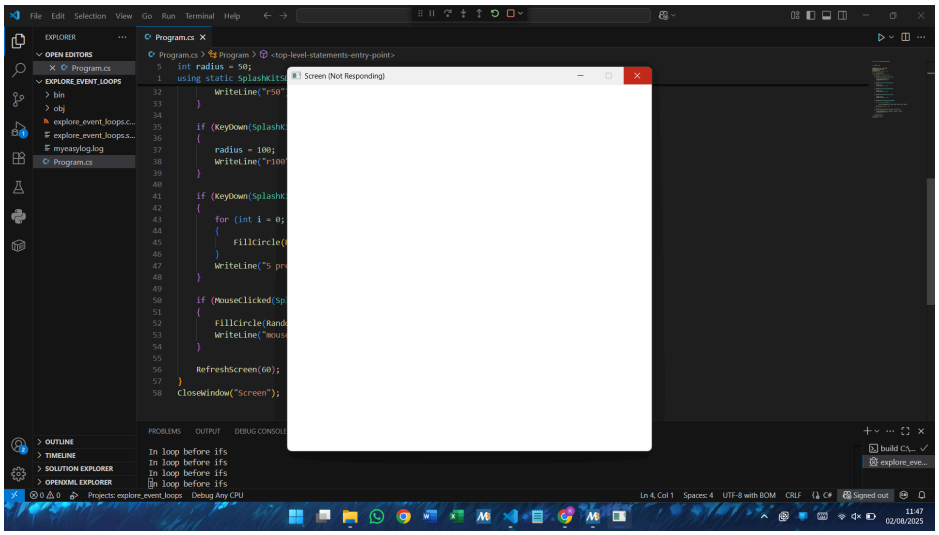
```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                dotnet

Path to file: C:/Users/Vedant/Desktop/mycode/my_c#_project/music_list/817843__josefpres__piano-loops-158-efect-4-octave-long

Loading C:/Users/Vedant/Desktop/mycode/my_c#_project/music_list/817843__josefpres__piano-loops-158-efect-4-octave-long failed!

1: Load song
2: Play song
3: Stop song
4: Quit
Option: 1


What is the name of the song: test
Path to file: C:/Users/Vedant/Desktop/mycode/my_c#_project/music_list/817843__josefpres__piano-loops-158-efect-4-octave-long-loop-120-bpm.wav

Loading C:/Users/Vedant/Desktop/mycode/my_c#_project/music_list/817843__josefpres__piano-loops-158-efect-4-octave-long-loop-120-bpm.wav Passed!

1: Load song
```

```
{
    case "1":
        WriteLine();
        Write("What is the name of the song: ");
        song = ReadLine();
        Write("Path to file: ");
        song_path = ReadLine();
        WriteLine();

        if (File.Exists(song_path))
        {
            LoadMusic(song, song_path);
            WriteLine($"Loading {song_path} Passed!");
        }
        else
        {
            WriteLine($"Loading {song_path} failed!");
        }
        WriteLine();

        break;
```

The loading {file path} passed or failed was a bit tricky as I wasn't familiar with the file.exists function. From my coding experience in visual basics, we used try-catch to verify if a file path was correct since an incorrect file path would crash the program. Here an incorrect file path does not crash the program instead output an error message, Thus thus try-catch in c# did not work.

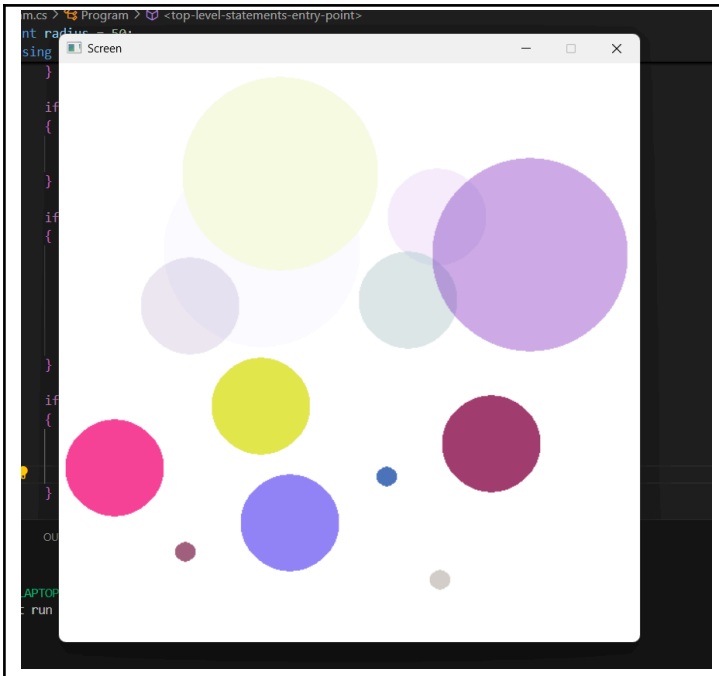
Beside that the program runs flawlessly.

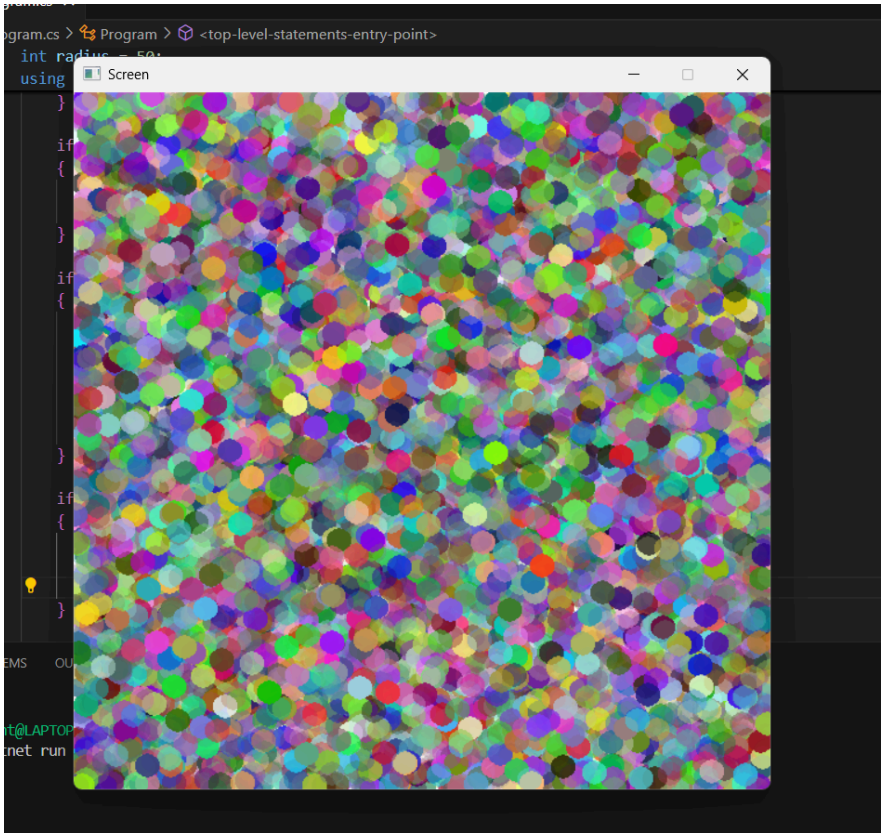
02/08/2025->03/08/2025 - Coded Explore event loops; test your knowledge



Program.cs > Program > <top-level-statements-entry-point>

```
int radius = 50;
using static SplashKitSDK.SplashKit;
ClearScreen(ColorWhite());
RefreshScreen();
WriteLine("before loop"); // test

while (!QuitRequested())
{
        WriteLine("In loop before ifs"); //test

    if (KeyDown(SplashKitSDK.KeyCode.CKey))
    {
        ClearScreen(RandomColor());
        RefreshScreen();
        WriteLine("In c"); //test
    }

    if (KeyDown(SplashKitSDK.KeyCode.SKey))
    {
        radius = 10;
        WriteLine("r10"); //test
    }

    if (KeyDown(SplashKitSDK.KeyCode.MKey))
    {
        radius = 50;
        WriteLine("r50"); //test
    }

    if (KeyDown(SplashKitSDK.KeyCode.LKey))
    {
        radius = 100;
        WriteLine("r100"); //test
    }

    if (KeyDown(SplashKitSDK.KeyCode.Num5Key))
```





I had a problem where the screen wasn't responding, thus I added test output prompts, there I learned that I had forgotten to write "ProcessEvents();".
Once I added that the program works.

Through the process of coding this program, I learned and understood how the key presses function works.

Brief Summary of Concepts

Concept / Commands	Key Idea / Concept
Boolean Data	It can have two values; True or False.
Comparison	These expression allows us to compare two values and determine the truth of a given condition
Logical Operators	They are comparison operators which allows us to compare two values
And	It is represented as '&&' in C#. It compares two boolean values. For example, a && b, if both a and b are true, the result will be true. If either a or b are or even if both are false, the result will be false.
Or	It is represented as '||' in C#. It compares two boolean values. For example, a || b, if either a or b are true or even if both a and b are true, the result is true. if both a and n are false, then the result is false.
Not	it is represented as '!' in C#. It changes the 'state' of a boolean variable. For example, let's say I have a boolean variable, a, and it is true. !a will give false.
If statement	It allows us to selectively run code based on the value of a boolean expression called condition
Switch case statement	It allows us to create paths that execute code based on the value of an expression.
While loop	It is a pre-condition loop that continues to iterate until the condition becomes false.
Do while loop	It is a post-condition loop that continues to iterate until the stopping condition becomes true.
For loop	It is a loop construct that continues to iterate for a set number of times.
Break	It is a statement that is used to jump out of a current loop.
Continue	The continue statement is used to jump back to the condition of the current loop.
Indentation	INDENTATION is very important. It makes the program more readable and easy to understand. It works as follows; for each block of code, a different amount of receding space is added.
Go to	It is a statement that allows us to jump anywhere in the code.

Reflection

What is the most important thing you learned from this and why?

The most important things I learned are the multimedia methods like ProcessEvent(), QuitRequested(), keyDown() methods, and Rnd() and the control flow statements like if statement, case statement, the three loop statements as well as the jumping,break and continue statements. They are what I consider to be core methods that I will use in most programs that I will write.

Additionally, learning how to use the debugger in real time while coding the programs mentioned above in ‘Study to Acquire, Refresh, and/or Extend Capability” was an invaluable learning experience.

How do you feel sequence and control flow are connected? Does understanding sequence help you understand control flow?

Sequence is like a straight line, continuously moving forward while control flow is like branches to that line going out from the mainline and may rejoin the line at different points. Control flow can go backwards on the mainline(loops) as well as providing alternate branches for the code execution to follow(selection statements).

Why is indentation so important?

Indentation makes code more readable and easier to understand. Imagine reading a book without any paragraphs, from start to finish, it is just line after line, it would be difficult to read, right? For coding it is the same, without indentation, the line would be confusing to read and understand what is. Indentation helps facilitate that by adding preceding spaces before each block of code where a block can be a construct like selection.

References

Generative AI acknowledgement

No AI was used for either the code or for this document.